

FlexRound: Learnable Rounding based on Element-wise Division for Post-Training Quantization

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Abstract

Post-training quantization (PTQ) has been gaining popularity for the deployment of deep neural networks on resource-limited devices since unlike quantization-aware training, neither a full training dataset nor end-to-end training is required at all. As PTQ schemes based on reconstructing each layer or block output turn out to be effective to enhance quantized model performance, recent works have developed algorithms to devise and learn a new weight-rounding scheme so as to better reconstruct each layer or block output. In this work, we propose a simple yet effective new weight-rounding mechanism for PTQ, coined *FlexRound*, based on element-wise division instead of typical element-wise addition such that FlexRound enables jointly learning a common quantization grid size as well as a different scale for each pre-trained weight. Thanks to the reciprocal rule of derivatives induced by element-wise division, FlexRound is inherently able to exploit pre-trained weights when updating their corresponding scales, and thus, flexibly quantize pre-trained weights depending on their magnitudes. We empirically validate the efficacy of FlexRound on a wide range of models and tasks. To the best of our knowledge, our work is the first to carry out comprehensive experiments on not only image classification and natural language understanding but also natural language generation. Moreover, we demonstrate, for the first time, that large language models can be efficiently quantized, with only a negligible impact on performance compared to half-precision baselines, achieved by reconstructing the output in a block-by-block manner. Our code is available at https://github.com/onliwad101/FlexRound_LRQ.

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1. Introduction

In recent years, deep neural networks have achieved unprecedented success across a wide variety of domains such as computer vision, natural language processing, and automatic speech recognition. Unfortunately, as these networks continue to improve and surpass human-level performance, the computational resources and memory usage required also increases as the architecture becomes more complex. To reduce the model size and accelerate inference operations, many researchers have attempted diverse compression techniques such as network quantization (Courbariaux et al., 2016) and network pruning (Han et al., 2016). In this paper, we concentrate on network quantization due to the advantage that INT4 or INT8 quantization allows us to accelerate quantized neural networks using off-the-shelf accelerators such as the NVIDIA A100 Tensor Core GPU (Wu et al., 2020) or ARM Cortex MCUs (Kim et al., 2021).

Network quantization techniques can be broadly divided into two categories: quantization-aware training (QAT) and post-training quantization (PTQ). QAT is a method where the quantization of the networks is incorporated during the training process, as proposed by various research works such as Jung et al. (2019); Jain et al. (2019); Zhao et al. (2020); Esser et al. (2020); Lee et al. (2021). We note that QAT results in a marginal performance difference between the full-precision and quantized versions of the neural network. Yet, QAT requires end-to-end retraining or fine-tuning on a full training dataset, which often causes an enormous amount of time and resources to obtain a quantized neural network with competitive performance. Furthermore, a whole training dataset may not be available due to data privacy issues or demands to utilize legacy models. Such drawbacks of QAT are the reasons why researchers recently pay more attention to PTQ (Zhao et al., 2019; Wang et al., 2020; Nahshan et al., 2021) that needs neither a full training dataset nor end-to-end learning at all.

PTQ had been initially performed via rounding-to-nearest by minimizing the quantization error in the parameter space. However, this approach suffers from severe performance degradation. Since it is reported that the loss degradation resulting from quantization can be approximated as the second-order error in Taylor Expansion by viewing quan-